

Reducing Redundancy in Vietnamese Multi-Document News Summarization with Position-Aware LexRank and MMR

COMP5040 – Natural Language Processing | VinUniversity Spring 2026

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Demo: <https://nlp-final-project.streamlit.app>

GitHub: https://github.com/cnvcuong/VinUni_Spring26_NLP_COMP5040_FinalProject_Group1

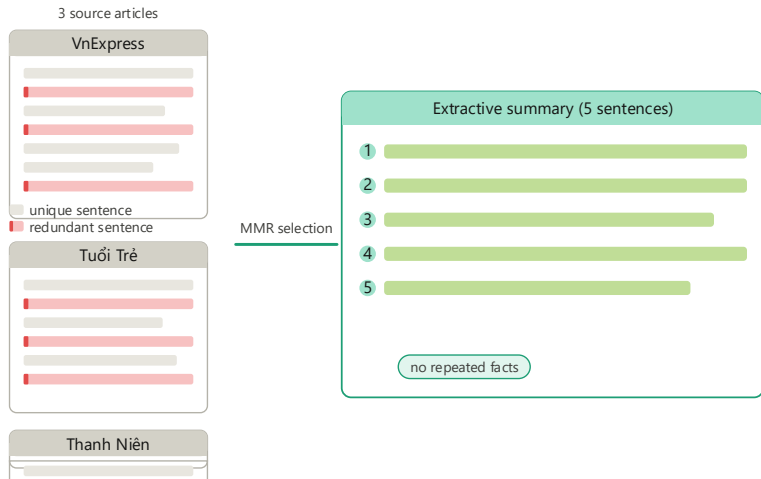
Why Does This Problem Matter?

Multiple Vietnamese news outlets often publish near-identical reports for the same event. Readers and downstream systems waste time processing repeated facts; manual summarization cannot scale. Extractive multi-document summarization aims to produce concise, faithful summaries that cover the event while minimizing redundancy.

01

Task definition

Input: a cluster of Vietnamese articles about one event. **Output:** 5-sentence extractive summary that maximizes factual coverage and minimizes redundancy.



02

Core challenges

- Saliency — identify the most informative sentences
- Redundancy — avoid selecting repeated facts across sources

03

Research questions

- Does LexRank outperform Lead-k on Vietnamese clusters?
- Does a position-and-title prior improve LexRank?
- Can MMR reduce redundancy without hurting ROUGE?
- Do BERT embeddings outperform TF-IDF here?

ViMs — Vietnamese Multi-Document Summarization Dataset

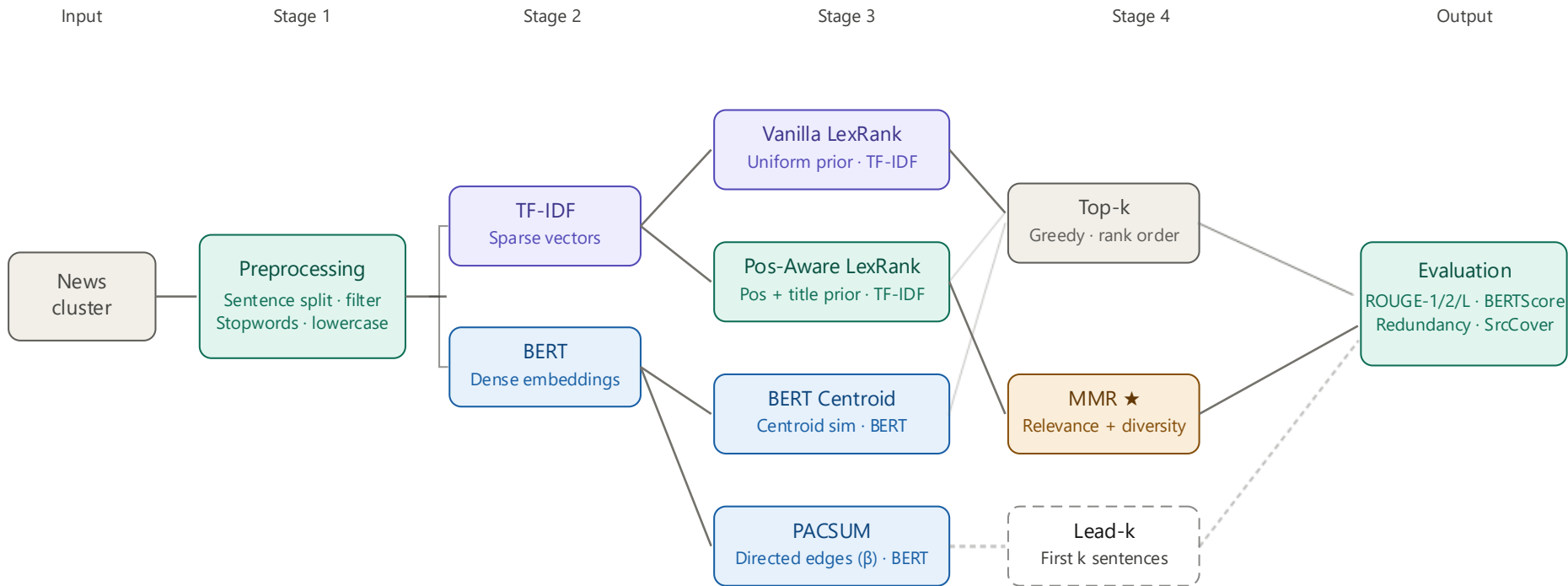
The ViMs dataset (CLC-HCMUS) contains clustered Vietnamese news events designed to study redundancy across multi-source reporting.

Statistic	Value
Event clusters	300
Source articles	1,945
Min articles per cluster	4
Max articles per cluster	10
Avg. articles per cluster	6.48
Reference summaries per cluster	2
Total candidate sentences	24,912
Max system summary length	5 sentences

❑ **Key characteristic:**
many near-duplicate sentences across outlets – redundancy is the primary obstacle to concise coverage.

End-to-End Summarization Pipeline

We implement a modular pipeline so experiments isolate representation, centrality, and selection choices.



Six Summarization Methods



Lead-k

Baseline: first k sentences from earliest article (no centrality).



Vanilla LexRank

TF-IDF similarity graph, uniform teleport prior, top-k selection.



Position-Aware LexRank

TF-IDF graph with position + title teleport prior, top-k.



BERT Centroid

Dense embeddings averaged per cluster; sentences ranked by centroid similarity.



PACSUM

Directed BERT graph (β controls positional bias), top-k selection.



Position-Aware LexRank + MMR ★

Proposed: TF-IDF LexRank with pos+title prior, MMR iterative selection to reduce redundancy.

Representation tracks: TF-IDF (fast, interpretable) vs. BERT (semantic, denser similarity).

How the Proposed Method Works

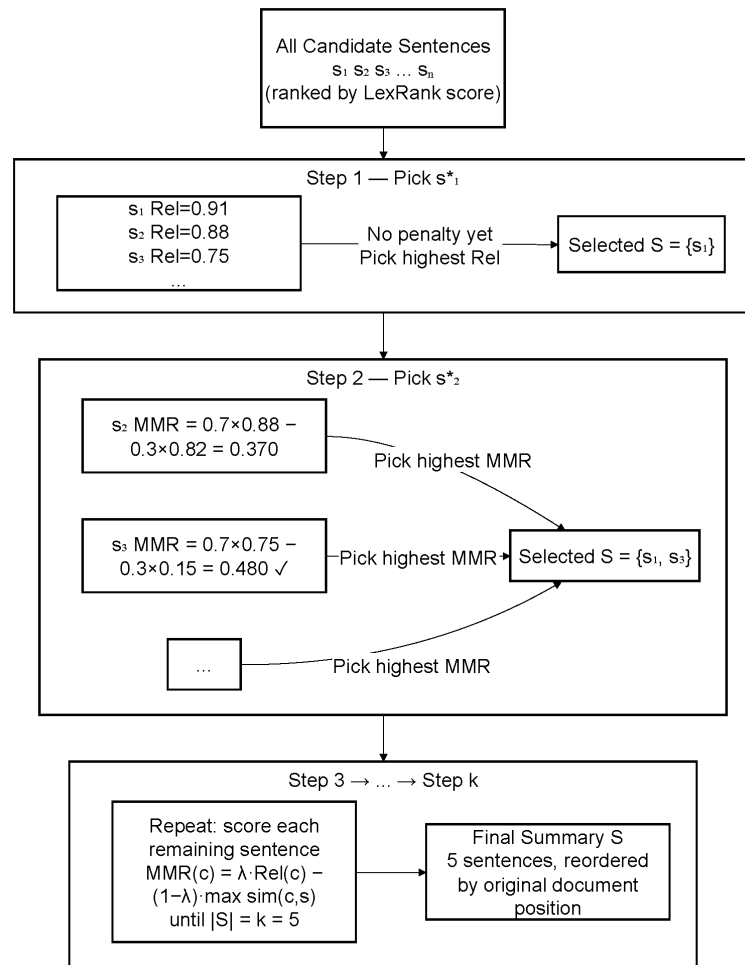
- **Step 1:** No sentences selected yet
→ pick sentence with highest LexRank score
- **Step 2:** Penalise candidates similar to selected
→ novel sentence wins despite lower relevance
- **Step 3:** Repeat — each new sentence must add new information vs. all selected so far

Formula:

$$\text{MMR}(c) = \lambda \cdot \text{Rel}(c) - (1-\lambda) \cdot \max \text{sim}(c, s)$$

$\lambda = 0.7 \rightarrow 70\% \text{ relevance, } 30\% \text{ diversity}$

❏ **MMR does not trade off ROUGE for diversity.** On this dataset, reducing redundancy **frees up** summary slots
→ ROUGE improves simultaneously.
Redundancy: 0.3369 → 0.2147 (-36.3%) **while**
ROUGE-L: 0.2922 → 0.3044 (+4.2%)



Evaluation Setup

- Metrics

- ROUGE-1/2/L F1 — lexical overlap (unigram, bigram, LCS)
- BERTScore F1 — semantic soft alignment (bert-base-multilingual-cased)
- Redundancy — average pairwise cosine similarity of selected sentences (lower is better)
- SrcCover — fraction of source articles contributing at least one selected sentence (higher is better)

- Experimental setup

300 clusters, average scores over both gold references.

DEFAULT HYPERPARAMETER VALUES.

Parameter	Value
Summary length k	5
LexRank threshold τ	0.10
Damping factor d	0.85
Position weight α	0.80
MMR λ	0.70
PACSUM β	0.00
Min. sentence tokens	4
BERT model	<i>paraphrase-multilingual-MiniLM-L12-v2</i>

Results — All Six Methods (Default Parameters)

Method	R-1 F1	R-2 F1	R-L F1	BS-F1	Redund.	SrcCover
Lead-k	0.4513	0.2731	0.2897	0.7363	0.0945	0.1756
Vanilla LexRank	0.4656	0.2840	0.2862	0.7366	0.3307	0.6022
Position-Aware LexRank	0.4690	0.2977	0.2922	0.7452	0.3369	0.6416
BERT Centroid	0.4474	0.2609	0.2656	0.7314	0.6933	0.5972
PACSUM	0.4524	0.2637	0.2705	0.7299	0.5176	0.2578
Pos-Aware LexRank + MMR ★	0.5135	0.3354	0.3044	0.7569	0.2147	0.6166

Takeaway 1

LexRank outperforms Lead-k on ROUGE and drastically increases SrcCover (0.18 → ~0.60).

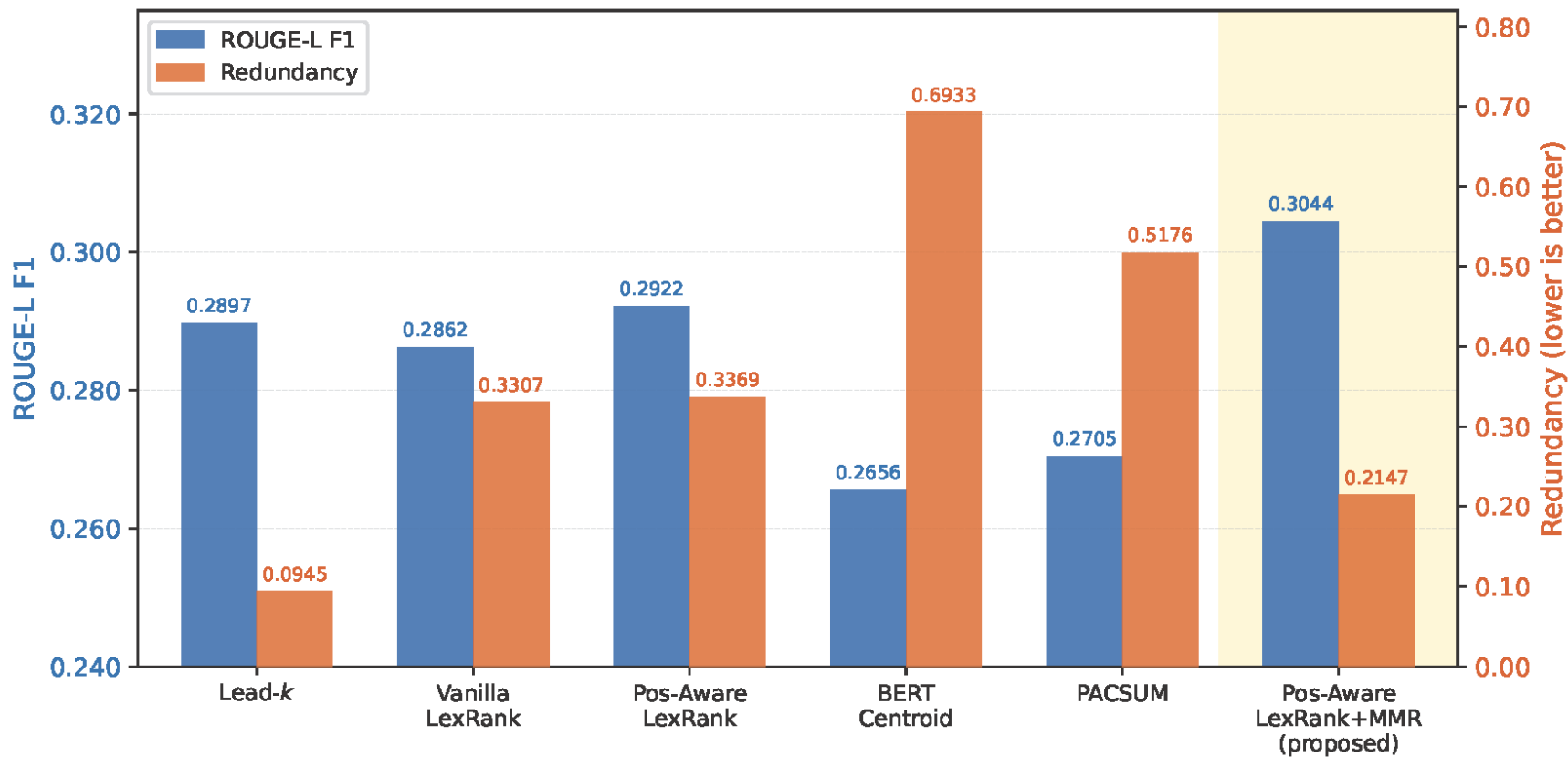
Takeaway 2

Position prior yields consistent ROUGE gains and better coverage (SrcCover 0.60 → 0.64).

Takeaway 3

Adding MMR reduces redundancy by ~36% while improving ROUGE-L (0.2922 → 0.3044).

Results — All Six Methods (Default Parameters)



Hyperparameter Grid Search — Key Findings

We ran an 80-configuration grid search over λ , τ , and α to optimize ROUGE-L. Search space:

- $\lambda \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$
- $\tau \in \{0.05, 0.10, 0.15, 0.20\}$
- $\alpha \in \{0.6, 0.7, 0.8, 0.9\}$

λ	τ	α	R-1	R-2	R-L
0.6	0.05	0.7	0.5300	0.3508	0.3140
0.6	0.05	0.6	0.5286	0.3478	0.3126
0.6	0.05	0.9	0.5272	0.3475	0.3119
0.6	0.05	0.8	0.5271	0.3470	0.3117
0.5	0.05	0.8	0.5304	0.3465	0.3108

Best configuration: $\lambda=0.6$, $\tau=0.05$, $\alpha=0.7 \rightarrow$ R-L = 0.3140 ($\approx +3.2\%$ over default). Observed patterns:

- Lower τ (denser graph) produces more stable centrality and better rankings.
- Lower λ (≤ 0.7) increases diversity weighting and consistently improves final summaries.

👉 Recommended default for deployment: $\lambda=0.6$, $\tau=0.05$, $\alpha=0.7$
(tradeoff: slightly more diversity with retained ROUGE performance).

BERT Methods: Why They Underperform TF-IDF

Empirical finding: BERT-based methods (BERT Centroid and PACSUM) underperform TF-IDF variants on ROUGE despite stronger semantic matching in some cases.



Dense surface overlap

Same-event clusters share dense, near-identical surface vocabulary; TF-IDF discriminates topical relevance effectively.



BERT increases redundancy

BERT links paraphrases and related sentences, producing denser similarity and higher redundancy (BERT Centroid redundancy = 0.6933).



PACSUM positional bias

PACSUM with $\beta=0$ concentrates on early sentences of few articles → low SrcCover (0.2578), missing broader coverage.

Method	BS-P	BS-R	BS-F1
Lead-k	0.7697	0.7061	0.7363
Vanilla LexRank	0.7767	0.7011	0.7366
Position-Aware LexRank	0.7859	0.7092	0.7452
Pos-Aware LexRank + MMR ★	0.7907	0.7265	0.7569
BERT Centroid	0.7663	0.7002	0.7314
PACSUM	0.7608	0.7019	0.7299

Conclusion: TF-IDF + MMR delivers both lexical and semantic gains on ViMs, better precision and controlled novelty.

Demo — Vietnamese News Summarizer

Live demo: <https://nlp-final-project.streamlit.app>

 **Model Parameters**

Method

Position-Aware LexRank + ...

Number of sentences (k)

5

Cosine threshold

0.10

Position weight

0.80

MMR Lambda (λ)

0.70

Recommended: Position-Aware LexRank + MMR with $\lambda=0.7$, threshold=0.1, position_weight=0.8

 **Vietnamese News Summarizer**

Position-Aware LexRank + MMR · Extractive Summarization

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Results, Conclusions, Limitations & Next Steps

Key empirical findings

- LexRank outperforms Lead-k, with better ROUGE ROUGE scores and much much higher source coverage.
- Position/title priors improve summary quality across ROUGE and coverage.
- MMR reduces redundancy while improving coverage and diversity.
- TF-IDF outperforms BERT on this corpus because same-event Vietnamese articles share strong keyword overlap.

Limitations

- Tokenization was regex-based; no Vietnamese word segmentation was used.
- ROUGE measures overlap, not factuality or fluency.
- The approach remains fully extractive, so it cannot compress or fuse information.

Future work

1. Add VnCoreNLP tokenization and re-evaluate the pipelines.
2. Fine-tune PhoBERT for better semantic matching.
3. Explore a hybrid extractive–abstractive approach and human evaluation.

Team & Q&A



Links

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Thank you — Questions?